● ALL SYSTEMS OK

Everything is Green Until It Is Not.

Metrics · Logs · Traces · The Gray Zone of AI Observability

From the trenches: Real stories, hard lessons, and the scars to prove it

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THE PLAN

The Agenda

01 The False Comfort of Green Dashboards

02 What to Actually Measure? and What does it cost you ?

03 Metrics, Logs, Traces — The Three Pillars

04 OTel — Vendor-Agnostic Observability

05 AI Monitoring — Welcome to the Gray Zone

07 Mind Map: Your Path Forward

SCENE: MONDAY MORNING

📊 Production Dashboard // Last 24h // Refreshed: just now

API Latency	Error Rate	Uptime
14ms ✅	0.001% ✅	99.99% ✅

errors 0.000

Because no alert was tied to **user success criteria or domain-specific SLIs (e.g., checkout success rate)**, nothing ever went red.

This is the classic *false comfort* — everything *looked* fine technically, but user-impacting failures *slipped through* undetected.

Let's Talk About The Three Pillars

METRICS

Numbers over time.
Cardinality traps await.

EARLY WARNING SYSTEM

LOGS

Text events.
Structure them or suffer.

CONTEXT ENGINE

TRACES

Distributed requests.
Where did 4s go?

THE DETECTIVE

THE BUDGET QUESTION ?

How Much Should Observability Cost?

~1-15%

of Cost of goods sold

Large orgs around 10 %
Cost conscious orgs around 5% or less
Ref: Steve Flanders Book (mastering Otel and Observability)

WHAT TO ACTUALLY MEASURE

RED + USE: Two Mental Models

RED — for SERVICES

R — Rate

Requests per second hitting the service

E — Errors

Fraction of requests that are failing

D — Duration

How long requests take → use p99, not avg

→ Use for: APIs, microservices, endpoints

USE — for RESOURCES

U — Utilization

% of time the resource is busy

S — Saturation

How much extra work is queued/waiting

E — Errors

Device-level errors: disk, packets dropped

→ Use for: CPUs, disks, DBs, queues, network

RED = Is my service healthy? **USE = Is my infrastructure healthy?**

Cardinality Will Destroy You

❌ CARDINALITY BOMB

```
http_requests_total{
  user_id="{uuid}",
  request_id="{uuid}",
  session_id="{uuid}"
}
```

✅ LOW CARDINALITY

```
http_requests_total{
  method="POST",
  route="/api/users",
  status="200"
}
```

⚠️ **High cardinality = your TSDB goes bankrupt. Prometheus doesn't forget.**

avg(response_time) = 12ms ✅

⬇️ *then you look at a histogram...*

p50 = 8ms p90 = 42ms 🔥 **p99 = 4,800ms**

Averages are liars. Always use percentiles.

Structure or Suffer

❌ UNSTRUCTURED — enjoy your grep session

```
[2024-01-15 14:32:01] ERROR Failed to process request for user abc123 - timeout
after 30000ms, retry 3/3
```

✅ STRUCTURED JSON — queryable, alertable, loveable

```
{ "ts": "2024-01-15T14:32:01Z",  "level": "error",
  "msg": "request_failed",
"user_id": "abc123",  "retry": 3 }
```

💡 **Structured logs = O(1) insight. Unstructured = O(pain).**

4 Seconds Went Somewhere.

Traces tell you exactly where the time went. payments-svc owes you 4 seconds.

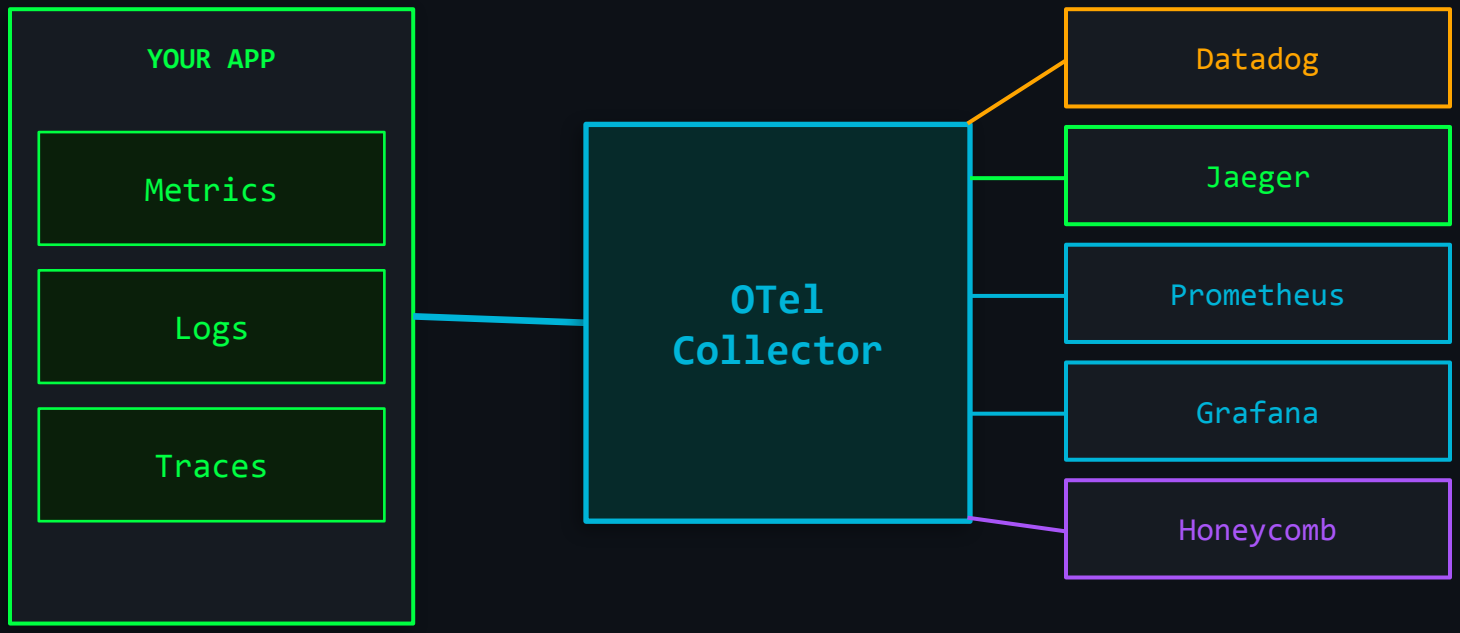
OPENTELEMETRY

Instrument Once. Ship Anywhere.

```
Before OTel: change vendor → rewrite ALL instrumentation
After OTel:  change vendor → update one config line
```

1 SDK

∞ backends



No more vendor lock-in. Swap backends without touching your app code.

NOW ADD AI TO YOUR STACK

Green → Gray

Everything you knew still applies.
But the failure modes are different.

```
Before: cpu.usage > 90%  →  📟 PAGE
Now:    LLM is... confidently wrong? Quietly drifting?
```

AI OBSERVABILITY

New Signals for a New Problem

🎭 Hallucination Rate

Model invents facts confidently.
No exception. No 500.

🌊 Relevance Drift

Output quality silently degrades
as context shifts.

💸 Token Cost Explosion

Input cardinality = $$$.
Sound familiar?

🔄 Context Abuse

128k tokens 'just in case'.
Latency goes brr.

🎯 Semantic Correctness

Valid JSON, meaningless output.
How do you alert?

THE HARD QUESTION

```
Conventional:
Latency > 500ms → PagerDuty → 😱
```

```
LLM System:
Hallucination > ???%  →
¯\_(ツ)_/¯
```

The answer: LLM-as-Judge + embedding distance + human evals

Measure	Traditional	LLM / AI
Correctness	assert result == X	LLM-as-Judge score
Latency	p99 < 200ms	TTFT + tokens/sec
Cost	server $$$	token cost / query
Failure	HTTP 5xx	graceful degradation?
Drift	version pinned	RAG chunk freshness

AI OBSERVABILITY

Evals Are Your New Unit Tests

WHAT TO EVAL

- 🎯 Factual accuracy
- 🌀 Relevance to query
- ☠️ Toxicity / bias
- 🧱 Format adherence
- 🏃 Latency + token cost
- 🎭 Hallucination score

TOOLS

LangSmith	traces + evals
Arize Phoenix	OSS observability
OpenTelemetry	Gen AI semconv
RAGAS	RAG eval framework
Braintrust	eval + prompt mgmt

OTel now has Gen AI Semantic Conventions (1.0 spec). Ship it.

YOUR OBSERVABILITY MIND MAP

Start Here. Ship This. In This Order.

OBSERVABLE SYSTEM

1 — Instrument with OTel
Auto-instrumentation first

2 — Structure your logs
JSON + trace_id correlation

3 — Define RED metrics
Rate, Errors, Duration / service

4 — USE for infra
CPU, memory, disk, queues

5 — Add AI/LLM evals
Hallucination + drift

6 — SLOs over dashboards
Error budgets, burn rates

7 — Cardinality governance
Review labels quarterly

8 — Budget audit
Target ~15% of cloud spend

Foundation first → Mature later → Budget always. Green is earned, not assumed.